

Joshua Aney

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EDUCATION

Montana State University

Bachelor of Science, Computer Science, GPA: 3.96

Bozeman, MT

May 2026

SKILLS & COURSEWORK

Languages: Python, MATLAB, Java, C, SQL, C#

Frameworks: PyTorch, TensorFlow, WebDataset, .NET Core, SLURM, NumPy, Matplotlib

Tools: Linux, JetBrains Products, GitHub, VSCode, Adobe Illustrator, Overleaf, Claude Code

Coursework: Machine Learning, Linear Algebra, Artificial Intelligence, Greedy Algorithms, Flow Networks, Optimization, Linear and Integer Linear Programming, Technical Writing

EXPERIENCE

Undergraduate Research Assistant

HACR Lab (Harnessing Automation in Cybersecurity Reasoning), Montana State University

January 2026 - Current

Bozeman, MT

- **Tools:** Pytorch, Ghidra, High Performance Computing, SLURM
- Build pipelines to produce control flow graphs from compiled binary packages for code embedding generation.
- Design Graph Neural Network architectures using PyTorch based on recent research publications.
- Run machine learning workloads on Tempest, Montana State University's high performance computing system.

Undergraduate Research Intern

CaMP Lab (Cybernetics and Motor Physiology), Mayo Clinic

Summers 2024, 2025

Rochester, MN

- **Tools:** MATLAB, Signal Processing, Adobe Illustrator, Linear Algebra, Technical Writing
- Conducted algorithmic analysis on human SEEG data to study how the brain responds to auditory stimuli.
- Implemented signal processing pipelines including noise reduction, baseline correction, and artifact rejection.
- Applied and modified Canonical Response Parameterization to allow for better temporal feature extraction when applied to auditory evoked responses.
- Produced a First-author manuscript in preparation: **Aney J**, Jensen MA, Kerezoudis P, Baker MR, Miller KJ. *Parameterization of intracranial auditory evoked responses reveals cortical novelty and attention effects*. Mayo Clinic, 2025.

TECHNICAL PROJECTS

Machine Learning Models from Scratch | *Python, NumPy, Linear Algebra, Latex*

2024-2025

- Implemented the following machine learning algorithms from scratch: Naïve Bayes, K-Nearest Neighbors, K-Means, and DBSCAN, Multi-Layer-Perceptron, Q-Learning, SARSA, and Value Iteration
- Implemented the following optimization algorithms from scratch: Backpropagation, Genetic Algorithm, Differential Evolution, Particle Swarm Optimization, and Simulated Annealing

Bozeman Bike Trails Application | *C#, Razor Pages, SQL Server, Mapbox*

2024

- Developed a web application to track user-riden bike trail GPS data

Hand Tracking / ASL Alphabet Detection | *Python, TensorFlow, MediaPipe*

2024

- Created a pipeline to collect data, train a dense neural network, and provide real time predictions for video inputs.

EXTRACURRICULAR ACTIVITIES & LEADERSHIP

President

Montana State University Tennis Club

August 2023 – Present

Bozeman, MT

Section Leader

Montana State University Drumline

April 2025 – January 2026

Bozeman, MT